

EEHC DISTRIBUTION MATERIALS SPECIFICATION
SWITCHED POWER CAPACITOR BANK 1 K.V UP TO 24 K.V

EDMS 21- 401 - 1
Date: 23-01-2024

EDMS 21-401-2

TECHNICAL SPECIFICATION
FOR
SWITCHED POWER CAPACITOR BANK
1 K.V UP TO 24 K.V (INDOOR)

Issue: Jan. 2024 / Rev- 2

- This revision contains option items that must be selected by EDC before bidding.

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EEHC DISTRIBUTION MATERIALS SPECIFICATION**EDMS 21- 401 - 1****SWITCHED POWER CAPACITOR BANK 1 K.V UP TO 24 K.V****Date: 23-01-2024****1. SCOPE**

This specification covers the minimum requirements for switched Power factor capacitor bank more than 1k.v up to 24 kv

The design, engineering, manufacture, testing at the manufacturer's factory, painting, packing for transport, insuring, transportation by road and delivery at destination shall be supported.

2. Applicable standards

This specification shall be read in conjunction with latest revisions of the following EDMS specifications:

No.	Standard No.	Description
1.	EDMS 02-100 & EDMS 02-200	Alternating current disconnectors and earthing switches air type 12 kv Alternating current disconnectors and earthing switches air type 24 kv
2.	EDMS 02-103 & EDMS 02-203	Alternating current disconnectors and earthing switches SF6 type 12 kv Alternating current disconnectors and earthing switches SF6 type 24 kv
3.	EDMS 19-403	Pole mounted Air Disconnector Switch
4.	EDMS 19-400	M.V H.R.C. Fuses
5.	EDMS 11-400	M.V surge arrestors

Unless otherwise specified in this specification should be comply with the latest edition of IEC standard and should be designed, manufactured and tested in accordance to the latest applicable IEC standards as follows:

Table (1)

No.	Standard No.	Description
1.	IEC 60871-1	Shunt capacitor for AC power systems having a rated voltage above 1kv - Part 1:General
2.	IEC 60871-3	Shunt capacitors for AC power systems having a rated voltage above 1 k V - Part 3: Protection of shunt capacitors and shunt capacitor banks
3.	IEC 60871-4	Shunt capacitors for AC power systems having a rated voltage above 1 k V - Part 3: Internal fuses
4.	IEC 62271-1	High-voltage switchgear and control gear - Part 1: Common specifications
5.	IEC 62271-200	High-voltage switchgear and control gear Part 200: AC metal-enclosed switchgear and control gear for rated voltages above 1 kV and up to and including 52 kV
6.	IEC 62271-106	High-voltage switchgear and control gear - Part 106: Alternating current contactors, contactor-based controllers and motor-starters
7.	IEC 60529	Classification of degrees of protection provided by enclosures
8.	IEC 60076-1&6	Reactors
9.	ISO12944-2	Paints and varnishes — Corrosion protection of steel structures by protective paint systems- Part 2: Classification of environments

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The electrical and mechanical properties of the required switched capacitor bank should be guaranteed under the following environmental conditions:

Ambient temperature	-5°C to +45°C (50°C is an option according to --EDC requirements), (35°C monthly average of the hottest month, 25°C yearly average)
Maximum relative humidity	95 %
Altitude	Up to 1000 m above mean sea-level

4. GENERAL TECHNICAL DATA

The required capacitor bank should have a safe behavior and meet all requirements given below:

System highest voltage	3.6 kV	7.2 kV	12 kV	24 KV
Nominal operating (Service) voltage	3.3 kV	6.6 kV	11 kV	22 KV
Power frequency R.M.S. withstand voltage (1 min)	10Kv	20Kv	28 kV	50 KV
Impulse withstand voltage (1.2/50) μ s peak				
Indoor	20kV	40Kv	75 kV.	125 KV
outdoor	40 KV	60kv	95 KV	145 KV
Rated frequency	50 Hz.	50 Hz.	50 Hz.	50 Hz.
IP class for Enclosure	IP 41 (for indoor) IP 54 (for outdoor)			

5. CAPACITOR (SINGLE / THREE) PHASE:

1. The capacitor type shall be approved by EEHC (shall be its temperature range not less than D according table 1 of IEC 60871-1).
2. The capacitors should be delta / star connection, and its rate shall be as –EDC /customer requirements.
3. The capacitor unit shall be consists of.
 - a. Capacitor elements.
 - b. Internal fuses.
 - c. Internal discharge element.
 - d. Stainless container with two/ three bushing.

5.1. Ratings :

1. The KVAR effective value calculated at service voltage
2. The upper limit of operating temperature category of the required capacitor unit shall be 55 °C according to IEC 60871-1 the lower limit is -5 °C

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3. unit shall be suitable for operation voltages (according IEC 60871-1 clause 19.1 table 5) as follows:

100 % U_Ncontinuous

110 % U_NFor 12 hours every 24 hrs

115 % U_NFor 30 minutes in every 24 hrs

120 % U_NFor 5 min.

130 % U_NFor 1 min.

4. Capacitor unit should be suitable for continuous operation at R.M.S current of 1.3 times the current that occurs at rated sinusoidal voltage and rated frequency, excluding transients. Depending on the actual capacitance value, which may be a maximum of $1.1C_N$ The maximum current can reach of $1.43I_N$. these overcurrent factors are intended to take care of the combined effect of harmonics and over voltages up to and including 110 % U_N according to 19.1.

5. The capacitance tolerance shall be clause 7. 2 IEC 60871-1.

[The capacitance shall not differ from the rated capacitance by more than -5% to +10% for capacitor unit & 0% to +10% for banks, In the 3-ph units and banks the ratio of maximum to minimum values of capacitance between any two line terminals shall not exceed 1.05]

5.2. Capacitor (Elements):

- The dielectric material of a capacitor element shall consist of an all film material being suitable for the capacitors to operate on continuous load under the specified ambient conditions.
- The capacitor should be completely leakage proof. The impregnate should be according to IEC 60871 & IEC 60422 of a hydrocarbon type fluid characterized by high electrical strength and adequate physical and chemical properties and should be non-PCB (Poly-Chlorinated-Biphenyl), but rapidly biologically degradable, non-poisonous, of trouble-free disposal and (according to NFPA 30) have a flash point $>150^{\circ}\text{C}$. In order to ensure good ionization performance, the aluminum foils should be folded at the edge.

5.3. Internal Fuses :

- Each capacitor (element) shall have an internal fuse combined with a secondary solid foil electrode to ensure safe disconnection from the circuit until the end of its normal working life.
- Fuses should only rupture to protect the related capacitor unit from failure and should be capable of breaking flowing the current a failure of the capacitor unit without hazard.
- The ruptured current limiting fuse of each capacitor (element) shall withstand indefinitely the voltage imposed across it under all operating conditions.
- The remaining capacitor elements should be able to operate within the capacitor unit without disturbance for a present number of unit capacitors.

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- If one internal capacitor (element) fuse get burnt, this shall not affect the total power and capacitance of the concerned capacitor unit by more than 3% of the rated power and capacitance of the capacitor unit . During the design stage. contractor shall submit the detailed calculations for number of internal fuse element failure to be considered for safe operation-to distribution for approval. (during operation)
- Facilities should be provided to allow for safe simple and quick identification of defective capacitor units

5.4. Losses :W / KVAR:

- Losses in steady state rated voltage and frequency at 25 °C indoor ambient temperature should not be more than 0.2W/KVAR (Lower values are preferable).

5.5. Insulation Material:

1. The Insulation of the capacitor unit shall be of synthetic foils (poly propylene - film), however the tenderer can submit paper insulation as an alternative.
2. All PCB impregnates are not allowed for use in Egypt. Insulation shall be of the (Al - film). Type, each capacitor shall be completely vacuum impregnated with no voids or air pockets to prevent corona and permit mounting in any position.

5.6. Bushings:

- Capacitor shall be of the two / three bushings type. Bushing shall be of well process porcelain.
- Creepage distance shall be not less than 4.5cm/kv for outdoor installation, and shall be not less than 2.5cm / kv for indoor installation (**nominal voltage is reference**) .
- All bushings shall be hermetically soldered to the steel case. Heavy tin-plated clamp type, terminals suitable for standard copper conductors. All capacitor terminals shall be provided with insulating terminal cover

5.7. insulation Level :

The insulation level of the units should be equivalent to that of the bank otherwise extra-insulation to the units is necessary.

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- Each capacitor unit shall be equipped with internal discharge resistor which shall drain the residual voltage from the operating voltage to 75 Volts within 10 minutes after the capacitor is disconnected from the source of supply according to IEC publication. The maximum discharge time is 10 minutes for both capacitors and banks.

5.9. Capacitor (Element) Container:

The capacitor elements should be enclosed in a stainless steel housing with minimum thickness 2mm and all joints welded and tested for liquid tightness.

5.10. Connections:

The connections between individual capacitor units in the same block should be such that a failed capacitor unit can be easily replaced. All interconnections between various components should be preferably carried out (as distribution company requirement) with tinned copper bus bar/XLPE single core copper cable with suitable cross section area .

5.11. Marking:

Capacitor shall be has rating name plate of weather proof materials, fitted I a visible position, showing in all cases the appropriate items indicated below, the entries on the name plate shall be indelibly marked according to IEC 60871-1.

Manufacturer`s name – Manufacturing year – Serial No. – Rated KVAR – Capacitance in μF
 Rated Voltage – Rated frequency – Temperature Category – discharge device – insulation level
 – internal fuses - IEC – connection symbol.

6. REACTOR (Inrush Current limiting Coil)

- Reactors are required to limit the transient currents produced during the connection of capacitors should be not exceed the rated making current of LBS
- Current limiting (damping reactor)/detuning reactors should be connected in series with each unit to limit the inrush current to an acceptable value (according to IEC 60871) and reduce harmonics to an acceptable level (according to IEEE 519).
- The current which flows upon energizing should be declared and should take into account the contribution from parallel connected capacitors.

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4. The reactors for indoor application at rated voltage 24 KV and should be air - core reactors and should be provided with supporting insulators including base plate and accessories ready for installation
5. The reactor should be capable of withstanding the inrush currents, which can occur during the life of the capacitor unit.
6. The reactor shall be has highest mechanical and short circuit strength, and low noise levels.
7. Reactors can be dry type / air-core or gapped-core.

a. General technical Data as following table:

Item	Unit	12 KV	24kv
Rated Voltage U_r	(kv)	12	22
Power frequency withstand for 1 min. U_d	(kv)	28	50
Lightning impulse withstand U_p	(kv)	75	125
Short duration nominal current for 1 sec.	(A)	43 I_n	100 I_n
Dynamic Current	A	2.5 I_t	2.5 I_t
Ambient temperature category		Category D	

b. Marking

reactor shall be has rating name plate of weather proof materials, fitted on a visible position, showing in all cases the appropriate items indicated below, the entries on the name plate shall be indelibly marked.

Type of reactor - Outdoor/indoor application – IEC – Manufacturer`s name - Manufacturer`s serial number – year of manufacture – insulation level – No. of phases – rated frequency – highest voltage – rated continuous current – rated thermal short circuit current and duration-thermal class (for dry type) – total mass – type of cooling

7. M.V CONTACTOR / VACUUM SWITCH

1. Contactor Sf6 or vacuum fixed type.
2. Contactor prepared for controlling capacitive loads.
3. The contactor should be restriking free and two stages pressure switch (alarm- trip).
4. Contactor shall be designed for a large number of operations specifically the loads associated with motors and capacitors.

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Item	Unit	12 KV
Rated Voltage U_r	(kv)	12
Power frequency withstand for 1 min. U_d	(kv)	28
Lightning impulse withstand U_p	(kv)	75
Rated operational current I_e	(A)	400
Utilization Category		AC4
Rated Short circuit breaking current (Isc)	kA	≥ 6
Rated Short time withstand current for (1sec.)	KA	≥ 6
Rated peak withstand current (I_p)	kA	≥ 15.6
Switch frequency Op./h		≥ 300
Mechanical endurance :		
For electromagnetic mechanism	Op.	≥ 1000000
For mechanical latch mechanism	Op.	≥ 50000
Closing time ms	m Sec.	40-145
Opening time (electrically latched) ms	m Sec.	15-40
Opening time (mechanically latched) ms	m Sec.	80-100
Operating temperature	$^{\circ}\text{C}$	40 $^{\circ}\text{C}$

5. for voltage more than 12kv can use C.B (**shall be able to break capacitive current of bank**) with protection relay instead of contactor and both of them shall be approved type by EEHC.

8. DESIGN CRITERIA

- The capacitor bank is of modulators type with suitable size to suit the available space, consists of two groups with total of (required MVAR) having rates of 1:1. or 1:2 The capacitor bank will be installed indoor, Taking into consideration the possibility of heavy polluted atmosphere and sharp variation of daily ambient temperature.
- The required automatic p.f controller for a shunt capacitor bank (two groups).
- Each group provided with a contactor /vacuum switch or M.V circuit breaker.
- Switching form step to step should be provided to operate automatically and manually.
- The operation sequence which could be carried out after closing the isolator and C.B is:
 - Put the selector on automatic position for both steps.
 - Measuring the reactive power of the load by the regulator (controller).

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c. If the power factor is lagging and reactive power =MVAR of step1+0.1MVAR,then :

- 1.Step1 contactor closes after time delay.
- 2.Measuring again the reactive power of the load.
- 3.If the compensation still not enough.
- 4.(Reactive power=MVAR of step2+0.1MVAR),then
- 5.Step2 contactor closes after time delay.
- 6.Measuring again the reactive power of the load.
- 7.If the P.F. becomes close to unity, both steps 1&2 remain in service.
- 8.If the P.F becomes leading then step should be switched off.
- 9.If the over voltage or over THD greater than 8% the contactors trip.

Note:

Delay time should be carried out by time delay relay and should be greater than capacitor discharge time.

6. Enclosure shall be include LBS (air type or SF6 type) with external fuse or C.B (shall be able to cut capacitive current of capacitor bank) for connection between capacitor bank , and load.
7. The capacitors shall be installed in such a way that there is adequate dissipation by radiation and convection of the heat produced by the capacitor losses. For indoor capacitor banks, one or more fans can be mounted in the bottom of the capacitor compartment to provide ventilation provided that enclosure temperature not exceed 50 C°.
8. Terminal blocks, internal wiring and cabling to the bank shall be included. The bidder shall provide the banks at the work with the necessary high voltage wiring of the capacitor units and other equipment and terminals of the bank.
9. Foundations, supporting structures and all electrical connections should be provided even if any item is not mentioned particularly in these specifications.
- 10.The MV capacitor units shall be complete in all respects necessary for their effective and trouble free operation when connected to the MV system.

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11. Enclosure should be made of hot rolled steel sheet 2 mm thickness at least before paint and should be treated against rust, and provided with primary and finishing coat of electrostatic painting powder of inside and outside, the external coat is preferred to be high gray color (RAL 7035).
12. The enclosure should be of electrostatic powder with thickness of 60:80 microns everywhere.
13. The enclosure should be galvanized steel to provide distinct corrosion protection, and finished with electrostatic paint **(as option according toEDC requirements).**
14. Degree of protection for the enclosure should be at least IP 41 for indoor installation.
15. The enclosure door should be provided with suitable gaskets, and should have locks with a master key.
16. The whole unit enclosure should have two eye-bolts diagonals to be lifted
17. The capacitor bank should be designed to give maximum reliability in service operation, inspection and maintenance.
18. Heater of 100-watt, 220V AC with a hygrostat and a M.C.B.
19. The capacitor bank should be completely assembled in one metallic cubicle (as drawing). Which contains: the high voltage connection, contactors, inrush current coils and auxiliaries.
20. The capacitor banks shall include the racks for the capacitor units and metal enclosure for live high voltage parts of the bank.
21. All doors shall be earthed by flexible wire not Y/G not less than 6 mm².
22. The enclosure door cannot be opened unless earth switch is in the "ON" position
23. The enclosure shall have a pocket or provision inside to store the instruction documents and other relevant information.
24. The enclosure should be has means of non-closing the doors during maintenance.
25. M.V voltage terminals for external connection between main panel and capacitor bank shall be supplied by the supplier.
26. In-case of Capacitor unit failure the contactor /C.B shall be open with sound alarm via horn.
27. The horn should be held by dual reset from push bottom and timer then a yellow lamp indicate the status of horn reset.

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28. Auxiliary voltage 220V or 110V DC/AC fed from main panel (**customer scope**).

29. Earthing B.B shall be not less than 150 mm²

30. One timer for controlling contactor operation

31. Capacitor bank shall be equipped the following protections

a. Unbalance current protection.

To protect the capacitor bank from being damaged due to unbalance current.

b. Over current & short circuit protection

c. Over voltage protection

d. Undervoltage protection

e. Low pressure of SF6 disconnector switch.

32. Capacitor bank shall be equipped with 3 metal-oxid surge arrestor on the incoming B.B and it shall be according to latest revision of EEHC specs. EDMS 11-400

33. Manufacturer shall supply data on expected lifetime for the capacitor bank(s) at rated service conditions.

9. TESTING**11.1 Type Test:**

All equipment shall be successfully type tested as EEHC approved laboratory in accordance to latest relevant standard for the following:

11.1.1 M.V. Capacitor according to IEC 60871-1

a. Thermal stability test

b. Measurement of (tan δ) of a capacitor.

c. AC test between terminals and container.

d. Lightning impulse voltage test between terminals and container.

e. Over voltage test.

f. Short circuit discharge test.

g. Disconnecting test on internal fuses.

h. Test of an external fuse in a combination with a capacitor.

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11.1.2 M.V contactor as per IEC 62271-106

- Mechanical endurance test.
- Making and breaking and reversibility test.
- Overload current withstand tests
- Short-circuit current making and breaking tests.
- Verification of operating limits and characteristics of overload relays.
- Change-over ability and reversibility, where applicable.
- Test of the striker mechanism.

11.1.3 Reactor as per IEC 60076-1&6

- Temperature rise test at rated continuous current.
- Lightning impulse test for current-limiting reactor.
- Measuring of power consumption of fans and oil pumps if any.

11.2 ROUTINE TEST :

Routine tests of the capacitor bank should be carried out according to latest relevant IEC standards by the manufacturer; a representative of –EDC will attend these tests before acceptance.

The --EDC have the right to add any test during delivery to check the quality of the product.

11.2.1 Visual inspection.

11.2.2 Mechanical operating tests.

11.2.3 Electrical operating tests shall be include.

- Capacitance measurement
- Capacitor losses tangent ($\tan\delta$) measurement.
- Voltage test between terminals.
- A.C. voltage test between terminal and container.
- Test of internal discharge device.
- Sealing test.

11.2.4 Function test of control circuit.

11.2.5 Dielectric (power frequency voltage withstand) test.

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- The tenderer should have a service center in Egypt covers all its deliverables.
- --EDC reserves its right to claim a free after sale service all over the warranty period guaranteed by the tenderer that covers all its deliverables, after sale services should include and not limited to the following:
 1. Technical support.
 2. Maintenance.
 3. Original Spare parts.
 4. Submittals.
- All the services, spare parts, materials and accessories will be needed during the after-sale maintenance should be guaranteed by the tenderer immediately and wherever required (Warehouse – Site – Lab.etc.)

11. GUARANTEE

- The supplier guarantees the capacitor bank against all defects arising out of faulty design or workmanship, or of defective material for period of 18 months from date of delivery or 12 months from date of operation.
- The good quality and manufacturing of the materials should be guaranteed.
- Any parts which on account of poor quality of the materials, manufacturing defects or poor assembly, should be replaced or repaired in the shortest time possible and free of charge.

12. MARKING

- The following data should be engraved or laser printed on an aluminum/stainless steel nameplate riveted on the capacitor bank and located in a visible and legible place when the switchboard is installed showing the following information:
 - Manufacturer's name
 - Type/Model
 - Serial Number and year of Manufacturing
 - Distribution company name
 - Order number
 - Rated KVAR
 - Rated voltage and rated insulation level
 - Rated frequency

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13. SUBMITTALS

a. Vendor shall provide the following with the quotation:

1. The inductance of each coil shall be calculated by the supplier and shall be submitted with offer for review and approval.
2. Maintenance, operation instructions and connection diagrams shall be submitted with offer.
3. Clause by clause compliance with the specification.
4. All necessary drawings including the scheme drawing of the power capacitor bank, dimensions, single line diagram of (control circuit, ...etc.)
5. Technical catalogues containing the technical data for all offered components of the unit should be submitted.
6. Installation and maintenance instructions.
7. Copy of type test report for capacitor, contactor, and reactor.
8. Submitted a valid approved letter from the Egyptian Electricity Holding Company if any.

b. Vendor shall provide the following after signing of purchase order:

1. Details of manufacturing and testing schedules
2. Routine test reports (including tightness test)

c. Vendor shall provide the following with delivering:

1. Operation (user) manual.
2. Handles, Master Keys, Locks..... for operation and locking.

14. TECHNICAL DATA SCHEDULE

- The guarantee schedules should be submitted and the tenderer should fill in the attached guarantee tables.
- --EDC reserves the right to reject any tender not accompanied with clear and complete guarantee tables.

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Capacitor Bank

No.	ITEM	EEHC Specs.				Supplier.	Remarks
1	Rated capacity (KVAR)						Natural and forced ventilation must be taken into consideration for the bank
2	Rated Voltage (kv)	3.6	7.2	12	24		
3	Max. Allowable over current	130 % I ₀					
4	Frequency	50 Hz					
5	Connection of capacitor units	Delta or Star					
6	Discharge time to 75V	600 sec.					
7	Insulation level of Bank KV	10/20 10/40	20/40 20/60	28/75 28/95	50/125 50/145		
8	Mounting	Indoor / outdoor					
9	Capacitor Bank According to	IEC 60871-1					

Table (2)
Over & Under Voltage Protection Relay

No.	ITEM	EEHC Specs.	Supplier Specs.	Remarks
1.	1 st Stage Voltage setting range Time range	100-115%U _n 0-5 Sec.		
2.	2 nd Stage Voltage setting range Time range	115-130%U _n 0-5 Sec.		
3.	Voltage setting range Time range	60%U _n 0-5 Sec.		
4.	Auxiliary Voltage	110-220 V DC or AC		
5.	Contact rating	7.5A – 250V DC/AC, NO		

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Table (3)
Inrush current coils (REACTOR)

No.	ITEM	EEHC Specs.				Supplier Specs.	Remarks
1	Inductance	X μ H					Calculated by the manufacturer inside the container
2	Max. Operating Voltage (kv)	3.6	7.2	12	24		
3	Service Voltage (kv)	3.3	6.6	11	22		
4	AC withstand voltage for 1 min.	10	20	28	50		
5	Impulse withstand voltage 1.2/50 μ s (kv)	20 or 40	40 or 60	75 or 95	125 or 145		
6	I _n (A)						
7	I _t (A)						

Table (4)
Contactor/ Vacuum switch or C.B for Switching Process of Capacitor Bank

No.	ITEM	EEHC Specs.	Supplier Specs.	Remarks
1	Rated MVAR			
2	System Voltage			
3	AC withstand voltage			
4	Impulse with stand voltage			
5	Rated normal current			
6	Rated S.C. Current			
7	Auxiliary Voltage			
9	Capacitor Bank According to			

We guarantee the data given above for the equipment offered

Signature: Date:.....